

Energy Nation Technical Note

Distinguishing Policy Actions from Institutional / Procedural Actions

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Author: Energy Nation Modeling Team

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1. Purpose

This technical note clarifies the conceptual and modeling distinction between policy actions and institutional or procedural actions within the Energy Nation simulation framework. The distinction is critical for determining whether a scenario should be modeled using the Government Override + Contracted Demand Runner (government_override_runner_colab_v2.py) or the Unified Overlay Runner (unified_overlay_runner_colab_revised_v2_patched.py).

2. Conceptual Overview

Dimension	Policy Action	Institutional / Procedural Action
Definition	A formal rule-setting or resource-allocation decision by government that changes incentives, mandates, or rights across a sector.	A reform in how institutions operate, altering review processes, timelines, or administrative governance.
Primary Effect	Changes economic conditions or market certainty.	Changes decision latency and approval duration (the 'time-to-FID' pathway).
Typical Instruments	Legislation, regulation, fiscal measure, investment credit, loan guarantee, offtake agreement, target, mandate.	Administrative directive, regulatory process redesign, agency coordination protocol, permitting timeline, one-window process.
Model Representation	Government Override Runner (α = override strength, β = contract	Unified Overlay Runner (time compression Δt and hazard multiplier λ).

strength, θ = interaction).

Model Output	$\Delta p, \Delta EV$ — probability and expected-value uplift from policy confidence or market pull.	$\Delta p, \Delta EV$ — probability and expected-value uplift from faster institutional throughput.
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3. Example: Major Projects Office Two-Year Cap

“The Major Projects Office is working towards transforming regulatory processes to ensure that major projects are reviewed within two years, from start to finish.” — MPO website

Interpretation:

- The regulation or directive establishing a two-year cap is a policy instrument.
- The substance—shortening approval duration to two years—is an institutional / procedural reform.
- Therefore, the modeled effect is procedural, producing a time-compression in project approvals.

Modeling Guidance:

If the main effect is faster permitting or review → use Unified Overlay Runner (MPO overlay).

If the main effect is stronger market signal or economic incentive → use Government Override Runner (α - β grid).

4. Integration in the Energy Nation Stack

Risk Engine (Bayes + Cox) → Expected-Value / Developer Engine → Government Override + Contracted Demand Runner → Unified Overlay Runner → Portfolio Synthesis & Data Briefs.

C represents policy levers (mandates, contracts, investment signals). D represents institutional levers (time compression, procedural efficiency). Both feed into the same Δp / ΔEV aggregation layer.

5. Modeling Example

Scenario	α (Override)	β (Contract)	Time Compression (Δt years)	Hazard Multiplier (λ)	Runner	Interpretation
Baseline	0.00	0.00	0	1.00	–	Status quo
Mandate (Policy)	0.75	0.25	0	1.00	Government Override Runner	Strong policy signal

MPO Cap (Procedural)	0.00	0.00	-2	1.10	Unified Overlay Runner	Institutional acceleration
Transformative (Combined)	1.00	0.50	-2	1.10	Both (stacked)	Policy + procedural synergy

6. Summary Statement

In Energy Nation modeling, policy actions change what governments decide and fund; institutional or procedural actions change how and how fast governments decide. A regulation establishing a two-year cap on major-project reviews is therefore policy in form but procedural in effect, and should be modeled as a time-compression overlay rather than a market-incentive shift.

7. Recommended Citation

Energy Nation (2025). “Distinguishing Policy and Procedural Actions in Major Project Modeling.” Technical Note No. 2025-10-15. Energy Nation Modeling Repository.